

Brain Tumor Detection

Introduction

Problem Statement:

Manual interpretation of MRI scans is very time-consuming, and often prone to human errors, and requires highly trained radiologists. With the increasing amount of medical data, a need for automated tools which can help with these interpretations has never been higher.

Objective:

- 1) To implement a machine learning model in Python on Google Colab.
 - 2) To preprocess and augment MRI data for better model generalization.
 - 3) To achieve a classification accuracy exceeding 90%.
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Methodology

Dataset:

The project utilizes the **Brain Tumor MRI Dataset** sourced from Kaggle.

- 1) Total Classes: 4 (Glioma, Meningioma, Pituitary, No Tumor).
- 2) Data Structure: The dataset is organized into Training and Testing directories.
- 3) Data Acquisition: Data is downloaded using the Kaggle API (kaggle.json) directly into the Colab environment.

Tools and Technologies:

- 1) Language: Python 3.x
- 2) Frameworks: TensorFlow, Keras
- 3) Environment: Google Colab (leveraging GPU acceleration)
- 4) Libraries: NumPy, Matplotlib, OS, Zipfile

Data Preprocessing:

To ensure the model learns effectively, the following preprocessing steps were applied:

- 1) The images are resized to 224 x 224 pixels to match the input requirements of the MobileNetV2 model.
- 2) To avoid overfitting, the training data was artificially expanded by using random horizontal flips, random zooming(up to 10%) and random rotations(up to 10%)

System Architecture

Transfer Learning:

- 1) Base Model: MobileNetV2, pre-trained on the massive ImageNet dataset.
- 2) Strategy: The "base" of the model was frozen (weights were not updated), allowing us to leverage its pre-learned feature extraction capabilities.
- 3) Classification Head: A new, trainable dense layer was added on top to classify the 4 specific tumor types.

Implementation Details

The training process involved the following steps:

1. Environment Setup: Installing Kaggle libraries and mounting the dataset.
2. Pipeline Creation: Using `tf.keras.utils.image_dataset_from_directory` for efficient data loading.
3. Training: The model was trained for 20-30 epochs.

Results and Discussion

Performance Metrics:

The model was evaluated based on Accuracy and Loss.

Model - MobileNetV2

Training Accuracy = ~98%

Testing Accuracy = ~92-95%

Observations - High generalization; robust performance.

Visualizations:

- 1) Training Curves: The Transfer Learning model showed a smooth decrease in loss and a steady increase in validation accuracy, converging faster than the custom model.
- 2) Predictions: Qualitative analysis of the test set showed that the model correctly identified complex tumor structures that the custom CNN often missed.

Conclusion

This project successfully demonstrated that Deep Learning can classify brain tumors with high accuracy. The use of Data Augmentation and Learning Rate Scheduling were crucial in preventing overfitting and boosting performance.

Future Scope

- 1) Deployment: The model can be deployed as a web application (using Streamlit or Flask) for real-time usage.
- 2) Explainability: Implementing Grad-CAM to visualize where the model is looking in the brain scan to provide doctors with visual evidence.
- 3) 3D MRI: Expanding the model to process 3D volumetric MRI data instead of 2D slices.

References

- 1) Kaggle Brain Tumor MRI Dataset.
- 2) TensorFlow and Keras Documentation.
- 3) Sandler, M., et al. "MobileNetV2: Inverted Residuals and Linear Bottlenecks." CVPR 2018.